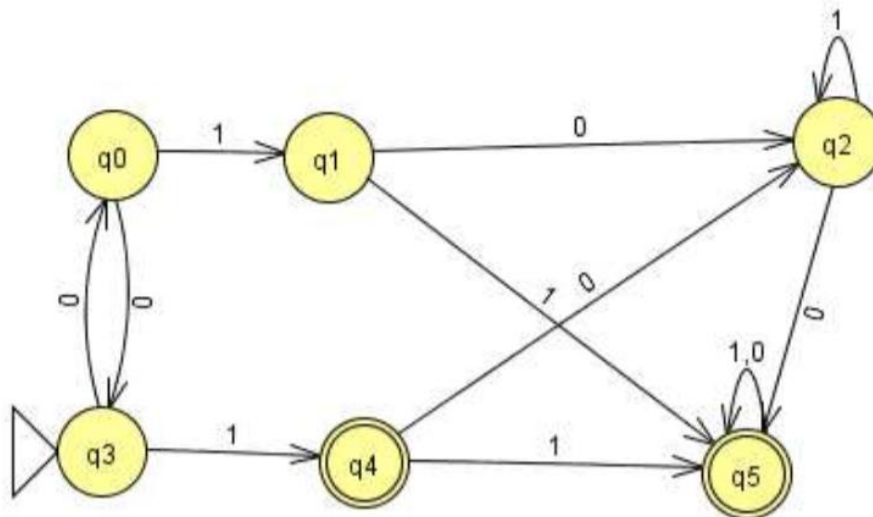


Example-1

4) Minimize the number of states of the below DFA. (10 Points)



Solution

DFA in table

	0	1
q_0	q_3	q_1
q_1	q_2	q_5
q_2	q_5	q_2
q_3	q_0	q_4
q_4	q_2	q_5
q_5	q_5	q_5

Applying Algorithm

①. $\{q_0, q_1, q_2, q_3\}$, $\{q_4, q_5\}$
Non-final states Final state.

$\delta(q_0, 0) = q_3$ } same set

$\delta(q_1, 0) = q_2$

$\delta(q_0, 1) = q_1$ } different set

$\delta(q_1, 1) = q_5$

q_0, q_1 & q_2 are not equivalent.

Now, Let's see if q_2 is equivalent to either q_1 or q_0

$\delta(q_0, 0) = q_3$ } different set

$\delta(q_2, 0) = q_5$

q_0 & q_2 are Not Equivalent.

$\delta(q_1, 0) = q_2$ } different set

$\delta(q_2, 0) = q_5$

q_0, q_1 & q_2 are Not equivalent

Similarly, q_3 is not equivalent to q_1, q_2, q_0 .

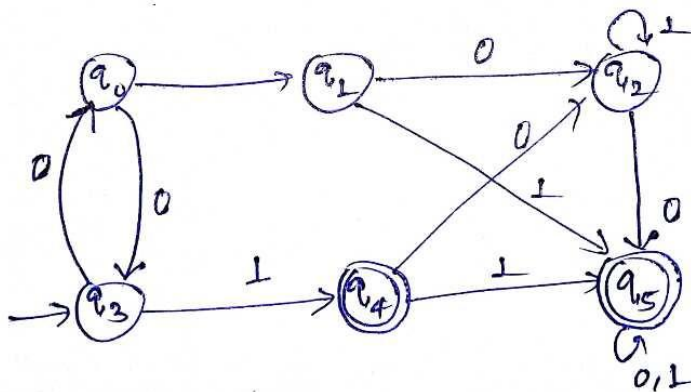
②. $\{q_0\}, \{q_1\}, \{q_2\}, \{q_3\}, \{q_4, q_5\}$

$\delta(q_4, 0) = q_2$
 $\delta(q_5, 0) = q_5$ } different set.

So, q_4 & q_5 are Not equivalent

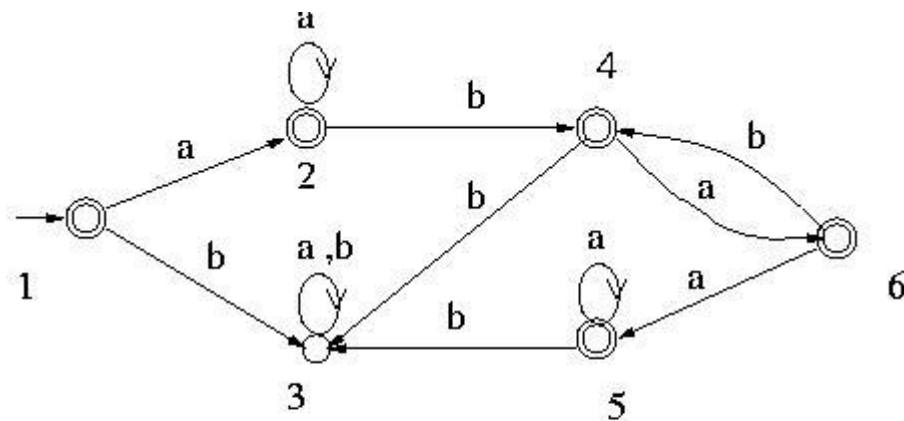
③. $\{q_0\}, \{q_1\}, \{q_2\}, \{q_3\}, \{q_4\}, \{q_5\}$

Minimized DFA:-



Example-2

Let us try to minimize the number of states of the following DFA.



DFA Example 2

Solution

Initially $\Pi = \{ \{ 3 \}, \{ 1, 2, 4, 5, 6 \} \}$.

By applying new_partition to this Π , $\Pi_{\text{new}} = \{ \{ 3 \}, \{ 1, 4, 5 \}, \{ 2, 6 \} \}$ is obtained.

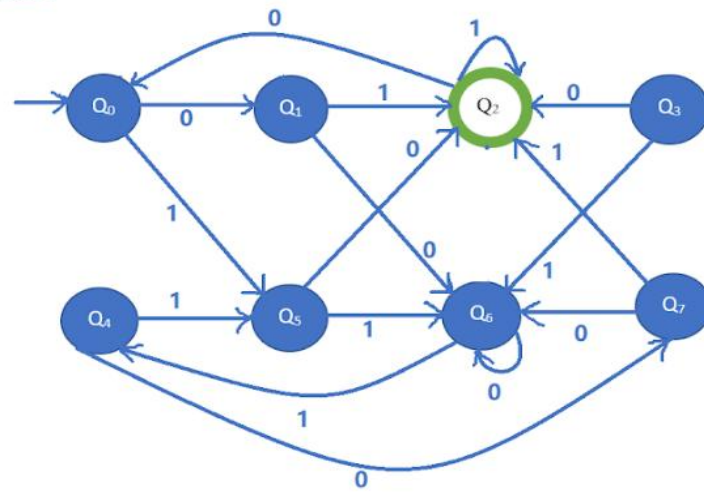
Applying new_partition to this Π , $\Pi_{\text{new}} = \{ \{ 3 \}, \{ 1, 4 \}, \{ 5 \}, \{ 2 \}, \{ 6 \} \}$ is obtained.

Applying new_partition again, $\Pi_{\text{new}} = \{ \{ 1 \}, \{ 2 \}, \{ 3 \}, \{ 4 \}, \{ 5 \}, \{ 6 \} \}$ is obtained.

Thus the number of states of the given DFA is already minimum and it can not be reduced any further.

Example-3

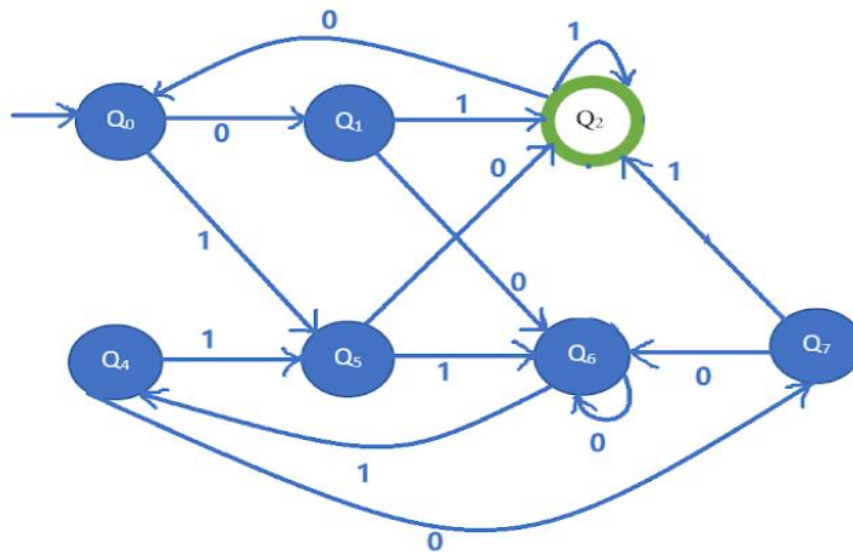
Example-2: Minimize the given DFA using partition method or construct minimized DFA using Equivalence method



Given DFA

Solution:

Step-1: Try to delete all the states to which we cannot reach from initial state



DFA without unreachable state

Step-2: Draw state transition table of DFA:

Present State	Next State	
	Input a	Input b
$\rightarrow Q_0$	Q_1	Q_5
Q_1	Q_6	$*Q_2$
$*Q_2$	Q_0	$*Q_2$
Q_4	Q_7	Q_5
Q_5	$*Q_2$	Q_6
Q_6	Q_6	Q_4
Q_7	Q_6	$*Q_2$

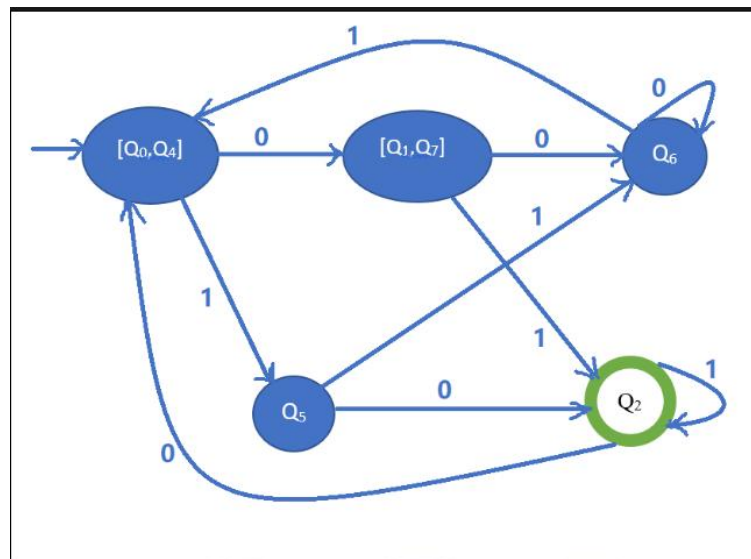
Step-3: Find out equivalent sets:

0-Equivalent Set: $[Q_0, Q_1, Q_4, Q_5, Q_6, Q_7] [Q_2]$

1-Equivalent Set: $[Q_0, Q_4, Q_6] [Q_1, Q_7] [Q_5] [Q_2]$

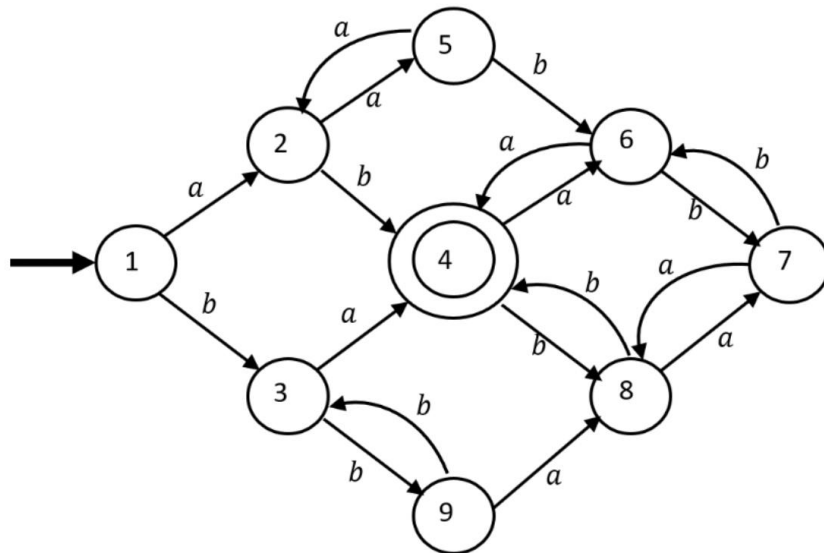
2-Equivalent Set: $[Q_0, Q_4] [Q_6] [Q_1, Q_7] [Q_5] [Q_2]$

3-Equivalent Set: $[Q_0, Q_4] [Q_6] [Q_1, Q_7] [Q_5] [Q_2]$



Example-4

Apply the DFA minimization algorithm to the DFA shown below. Show the minimized DFA, and construct a matrix indicating, for each distinguishable pair of states, the iteration in which it is checked off.



Solution

① Zero equivalence: {1, 2, 3, 5, 6, 7, 8, 9}

② (a) equivalence: {1, 5, 7, 9}, {2, 3, 6, 8}, {4}

③ (b) equivalence: {1, 5}, {7, 9}, {2, 8}, {3, 6}, {4}

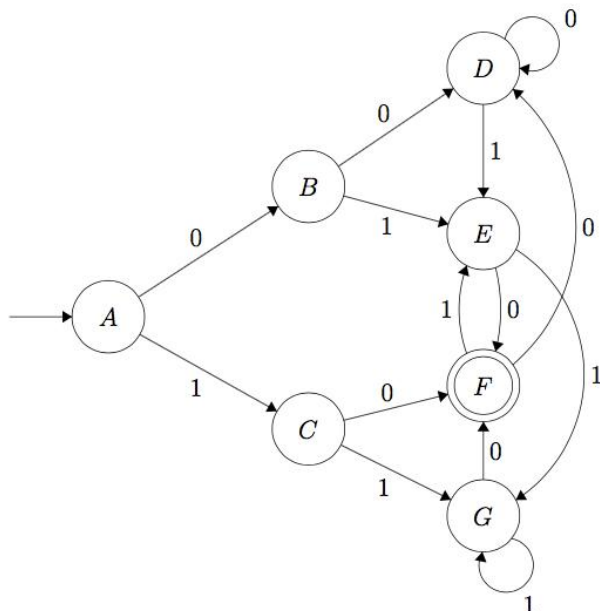
④ (c) equivalence: {1, 5}, {7, 9}, {2}, {8}, {3}, {6}, {4}

	a	b
1	2	3
2	5	4
3	4	9
4	6	8
5	2	6
6	4	7
7	8	6
8	7	4
9	8	3

	a	b
1,5	2	3
7,9	8	6
2	(1,5)	4
3	4	(7,9)
4	(6)	8
6	(4)	(7,9)
8	(7,9)	(4)

Example-5

Given the following DFA M. apply the DFA minimization algorithm and show the resulting minimized DFA. Show your graph or table and label each state in your minimal DFA with the set of states of M it corresponds to.



Solution

Step 1:
 All the states are divided into two partitions.
 final states & non-final states. Say iteration 0... P_0

Step 2:
 For each iteration, say P_0 , we check if the states goes to another state & if they are in the same sets. if not do the partition.

Step 3: Repeat the partition & at the end make the distinguishable sets as new states of reduced DFA.

Here
Step 1: 'f' is final state, so
 $P_0 = \{A, B, C, D, E, F, G\} \{F\}$
 Now we do partition say P_0 as partition S, such that S is divided into subsets such that the two states

Step 2:
 Since 'C' on 0 goes to F & '1' goes to G, both are in different sets. E on 0 goes to F & '1' goes to G. G on 0 goes to F & '1' goes to G.
 We partition into set $\{C, E, G\}$ as they all on 0 goes to F & 1 goes to G.
 New partition $P_1 = \{A, B, D\} \{C, E, G\} \{F\}$

Step 3:

Consider B on 0 goes to D & B on 1 goes E.
D on 0 goes to D & D on 1 goes E so grouping

$\{A\}$ $\{B, D\}$ $\{C, E, G\}$ $\{F\}$.

Step 4:

The above sets are distinguishable, thus now create these subsets as new states. Minimized DFA as below.

